

### Quick Activity 1: School Suppliers Calculator

A notebook costs **P45** and a pen costs **P12**. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

#### Inputs:

- Number of notebooks
- Number of pens
- Payment amount

#### Processes:

- Calculate cost of notebooks (notebooks x **P45**)
- Calculate cost of pens (pens x **P12**)
- Calculate total cost (notebook cost + pen cost)
- Calculate change (payment - total cost)

#### Outputs:

- Total cost
- Change

#### Tasks:

- 1.) Input number of Notebooks
- 2.) Input number of pens
- 3.) Input payment amount
- 4.) Compute notebook cost
- 5.) compute pen cost
- 6.) Compute total cost
- 7.) Compute Change
- 8.) Display total cost
- 9.) Display change

#### PseudoCode

START

```
    READ notebooks, pens, payment
    notebookCost ← notebooks × 45
    penCost ← pens × 12
    totalCost ← notebookCost + penCost
    change ← payment - totalCost
    DISPLAY totalCost
    DISPLAY change
```

END

## DRY RUN CALCULATIONS:

### Test Case 1:

Given: 3 notebooks, 2 pens, payment of ₱200

| Step | Variable     | Calculation | C    |
|------|--------------|-------------|------|
| 1    | notebookCost | 3x ₱45      | ₱135 |
| 2    | penCost      | 2x ₱12      | ₱24  |
| 3    | totalCost    | ₱135 + ₱24  | ₱159 |
| 4    | change       | ₱200 - ₱159 | ₱41  |

Total cost: ₱159.00

Change: ₱41.00

### Test Case 2:

Given: 2 notebooks, 3 pens, payment of ₱150

| Step | Variable     | Calculation | Result |
|------|--------------|-------------|--------|
| 1    | notebookCost | 2x ₱45      | ₱90    |
| 2    | penCost      | 3x ₱12      | ₱36    |
| 3    | totalCost    | ₱90 + ₱36   | ₱126   |
| 4    | change       | ₱150-126    | ₱24    |

Total cost: ₱126.00

Change: ₱24.00

Check your work:

- Are the quantities and payment accepted as inputs? YES
- Are the notebook and pen costs calculated separately before finding the total? YES
- Is the change calculated correctly? YES
- Are the total cost and change displayed? YES
- Can someone follow your steps without asking for clarification? YES, as long as the following task given is clearly understood and written any one can follow my steps.